

PORT ARANSAS 32 NNE, TX, USA

WMO: 747470

Lat: 28.3045N

Lon: 96.8230W

Elev: 15

StdP: 14.69

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 23906

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.9	32.3	20.8	15.6	37.4	24.5	18.6	41.7	12.4	59.0	10.8	57.0	1.3	N/A	0.311

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.6	93.0	78.5	91.6	78.8	90.5	78.8	81.4	88.4	80.8	87.7	80.3	87.1	5.3	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.7	154.3	84.4	79.1	151.0	84.3	78.4	147.7	84.1	44.9	88.7	44.3	87.9	43.8	87.4	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.2	8.8	8.1	DB	19.9	97.9	3.7	3.2	17.2	100.2	15.1	102.1	13.0	103.9	10.4	106.3	
			WB	19.1	82.5	4.1	0.9	16.1	83.2	13.7	83.7	11.4	84.2	8.4	84.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	70.3	53.5	59.0	64.5	70.4	76.7	82.0	82.6	82.9	79.9	72.1	62.9	56.8
	DBStd	12.47	9.36	9.79	8.38	6.54	5.16	2.86	2.28	2.49	3.79	7.85	9.66	9.56
	HDD50	168	73	31	8	0	0	0	0	0	0	1	15	40
	HDD65	1201	371	210	114	31	4	0	0	0	0	31	151	289
	CDD50	7589	181	284	458	612	827	959	1010	1020	898	687	403	250
	CDD65	3147	14	42	99	193	366	509	545	555	448	252	89	35
	CDH74	32500	40	98	284	988	3277	6000	6746	6906	4909	2594	548	110
	CDH80	11974	2	6	20	120	727	2348	2896	3184	1906	698	65	2
Wind	WSAvg	3.7	3.6	4.1	4.6	5.0	4.7	4.0	3.5	3.2	2.7	2.9	2.8	3.2
Precipitation	PrecAvg	37.6	2.6	2.2	2.8	2.4	3.6	2.8	4.1	3.2	5.2	4.0	2.6	2.4
	PrecMax	54.9	9.4	8.9	11.3	5.8	12.7	8.6	20.0	7.6	15.4	11.1	6.8	6.7
	PrecMin	16.0	0.0	0.1	0.0	0.5	0.0	0.0	0.0	0.6	1.0	0.0	0.0	0.6
	PrecStd	11.0	2.2	2.2	2.6	1.6	3.4	2.4	5.2	2.2	3.6	3.1	2.4	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.8	80.1	82.7	88.0	89.0	94.8	93.4	95.5	94.7	89.3	84.4	79.4
		MCWB	64.9	67.3	67.9	66.3	74.2	77.5	77.9	77.2	75.3	76.3	75.0	70.9
	2%	DB	74.7	77.0	79.2	82.8	86.9	91.0	91.9	93.2	91.3	87.5	82.0	77.2
		MCWB	66.1	69.1	69.6	71.7	76.4	78.0	78.7	79.2	77.9	77.4	73.8	69.4
	5%	DB	71.7	74.5	77.2	80.6	85.5	89.5	90.8	91.9	89.8	86.0	79.6	74.7
		MCWB	64.6	68.0	69.5	72.3	76.4	78.7	79.0	79.4	78.2	76.4	72.1	68.9
	10%	DB	69.1	71.8	75.1	79.0	84.1	88.4	89.7	90.8	88.4	84.1	77.2	72.0
		MCWB	64.0	67.3	68.7	71.7	75.9	78.7	79.1	79.4	78.1	74.9	71.7	68.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.4	72.4	73.2	77.3	79.6	81.6	82.0	82.4	81.7	80.3	77.6	73.7
		MCDB	73.4	77.3	78.4	80.8	84.6	87.6	89.0	89.7	88.2	85.3	81.7	76.3
	2%	WB	69.2	71.0	72.1	76.0	78.7	80.7	81.1	81.6	80.8	79.0	76.0	72.6
		MCDB	72.1	75.0	76.5	79.6	83.6	86.7	87.8	89.1	87.4	83.7	79.7	75.1
	5%	WB	67.7	69.9	71.2	74.9	78.1	80.2	80.4	80.9	80.1	78.1	74.4	70.8
		MCDB	70.3	73.1	75.0	78.5	82.8	86.3	87.2	88.3	86.7	83.2	78.2	73.3
	10%	WB	65.4	68.7	70.2	73.7	77.3	79.6	80.0	80.3	79.4	77.1	72.7	69.0
		MCDB	68.2	71.2	73.6	77.1	82.0	85.9	86.8	87.6	85.9	82.5	76.6	71.5
Mean Daily Temperature Range	5% DB	MDBR	21.7	18.4	16.6	16.1	13.9	13.7	15.0	16.6	16.5	20.6	21.1	20.1
		MCDBR	23.7	18.0	15.3	15.6	14.3	14.5	16.8	18.2	18.6	17.5	18.7	20.0
		MCWBR	15.8	11.6	8.3	8.6	6.6	5.7	6.8	6.9	8.1	9.6	11.7	14.1
	5% WB	MCDBR	16.3	14.6	13.6	11.1	9.9	11.6	14.1	15.4	14.0	15.9	15.9	18.0
		MCWBR	12.8	10.3	8.0	7.0	5.1	5.1	6.2	6.6	6.7	9.8	11.4	14.1
Clear-Sky Solar Irradiance	taub		0.357	0.367	0.384	0.437	0.482	0.456	0.477	0.454	0.444	0.392	0.380	0.346
	taud		2.488	2.463	2.409	2.267	2.173	2.309	2.266	2.333	2.323	2.461	2.443	2.534
	Ebn at Noon		280	287	287	274	260	266	260	266	265	274	269	279
	Edh at Noon		30	33	37	43	48	42	43	40	40	32	31	27
All-Sky Solar Radiation	RadAvg		938	1100	1444	1716	1941	2128	2087	1992	1656	1438	1079	849
	RadStd		104	179	139	159	131	120	186	109	145	148	117	108

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+202	+171

Nomenclature: See separate page